Astro Play

NASA Space Apps Cairo 2026 — Team Proposal

Challenge: Build a Junior Astronaut Mission Trainer

Executive Summary

Challenge selected: Build a Junior Astronaut Mission Trainer. Space content for kids is often too abstract to show the trade-offs behind a real mission, or too technical for a young learner. Astro Play closes this gap: a game, in Arabic and English, where kids aged 7–12 run a lunar or Martian outpost and discover that no choice is free.

Astro Play offers a 3D astronaut experience (Moon, Mars, orbital lab, Europa) with oxygen and tool-battery limits and an age-adaptive AI assistant built on retrieval-augmented generation (RAG) over NASA sources. For this challenge we will build an **Outpost Command** mode. Each sol, the player splits limited power between life support, radiation shielding, communications and science. Every option costs something: reinforcing shielding against a radiation event drains tonight's power.

The mode is driven by real NASA data. Radiation risk follows solar activity from NASA's DONKI space-weather database, and short fact cards use findings from the Open Science Data Repository (OSDR). Every source is labeled in the game, and archived data is marked as archived, never shown as live.

After each mission a debrief explains what happened and why. Events vary between playthroughs, so failing becomes a way to learn. The assistant explains terms such as radiation dose in plain language and declines questions beyond its sources.

We chose this challenge because it matches our belief that children learn science by making decisions and seeing the consequences.

Problem Definition

NASA is preparing a sustained presence on the Moon and journeys toward Mars, yet public understanding of what survival there requires remains thin. Astronauts must split limited power between life support, communications and science, weigh radiation exposure against mission goals, and live with resource costs that compound over time.

Space content for children usually fails in one of two ways. It stays abstract (posters and general facts) and hides the decision-making logic, or it assumes technical knowledge that a 7–12-year-old does not have. Entertainment-first games often lose accuracy. For Arabic-speaking children, quality content is rarer still.

The problem we address: how can real mission trade-offs, grounded in real NASA data, become tangible, accurate and fun for children aged 7–12, in their own language?

Background & Literature Review

Existing solutions. NASA publishes valuable open data, such as DONKI space-weather records and OSDR astronaut research, but these are expert-facing. Management games such as *Oxygen Not Included* and *Surviving Mars* make resource trade-offs engaging, but they target older players, are complex for children and do not use live NASA data or Arabic.

Research. A meta-analysis of serious games found they can improve learning and motivation compared with conventional instruction (Wouters et al., 2013). Learning through games works best when players act on the material, not only receive it (Gee, 2003). Retrieval-augmented generation grounds a language model's answers in a trusted document set and reduces invented answers (Lewis et al., 2020), which is vital for science education.

Gap. We found no tool that combines a child-friendly outpost trade-off simulation, real NASA space-weather and research data shown with clear attribution, a RAG assistant that stays within NASA sources, and Arabic-first design.

Methodology

Step-by-step process

- **Data selection:** use DONKI (solar flares, coronal mass ejections) for radiation risk, OSDR for astronaut health and biology findings, and NASA Moon/Mars facts for site conditions.
- **Trade-off model:** define three core resources (power, life support, radiation shielding) so that every choice has a cost, with communication delay added as events.
- **Build:** implement the Outpost Command loop on the Astro Play engine, with a clear dashboard for young players.
- **Data integration:** fetch DONKI through NASA's API with a labeled archived dataset as fallback; show sources in-game.
- **AI assistant:** extend the RAG knowledge base with outpost, radiation and OSDR topics, with age- and language-aware answers.
- **Testing:** automated tests, plus playtests with children aged 7–12 in Arabic and English.

Technologies

- **Python + Ursina Engine** (Panda3D) for gameplay and the dashboard
- **NASA open APIs** (DONKI) and datasets (OSDR) for real data
- **ChromaDB** vector search with keyword fallback, and **Google Gemini** for RAG answers
- **SQLite** for profiles, progress and achievements; **arabic-reshaper + python-bidi** for right-to-left Arabic
- **pytest** for testing

Fit to the challenge: the method targets the stated objectives directly: kids 7–12, competing resources, real NASA data, no cost-free choices, replayability and traceable sources.

Solution

How it tackles the challenge. In **Outpost Command**, the player runs a lunar or Martian outpost over a series of sols. Each sol they allocate limited power among life support, radiation shielding, communications and science, while watching oxygen and food. Examples of trade-offs: trading oxygen reserves for an extra research opportunity, or reinforcing shielding before a solar event at the cost of tonight's power. A "safe" option with no cost does not exist.

Real data, visible consequences. Radiation-event likelihood and severity follow recent solar activity from DONKI, so a shielding decision links to real space weather. OSDR findings appear as short fact cards. Every source is labeled in-game, and archived data is clearly marked as archived.

Replayability and learning. Events vary between playthroughs, and a debrief explains the cause of every outcome without punishing failure. The AI assistant explains terms like "radiation dose" and "sol" in context and nudges instead of giving answers.

Implementation steps: (1) trade-off model and balancing, (2) outpost loop and dashboard, (3) data integration and attribution, (4) assistant knowledge, (5) playtesting and polish. The prototype runs on desktop; a browser version is a next step.

Value Proposition

For children (7–12): they feel why mission planners cannot have everything. They learn by deciding, with plain-language explanations, a safe place to fail and a result that links back to real NASA science.

For parents and teachers: an Arabic-language tool that teaches real engineering trade-offs and data reasoning, with visible sources and an assistant that does not invent answers. Older siblings can use harder difficulty levels.

Why it beats alternatives

- **Versus posters and fact content:** players experience the decision logic, not just the outcome.
- **Versus commercial management games:** built for children, tied to real NASA data and clearly attributed.
- **Versus general chatbots:** answers come from NASA sources and adapt to age.
- **Versus English-only tools:** Arabic-first with correct right-to-left design.
- **Honest data:** live and archived data are always labeled.

Role of Team Members

Member	Role	Contribution
Basmala Saeed	AI Lead	Designs the RAG pipeline (ChromaDB + Gemini), DONKI/OSDR data integration, and age/language-adaptive prompting and safety limits.
Merhan Medhat	AI Lead & Presenter	Builds the assistant's knowledge base and behavior, checks the science for accuracy, and presents the demo.
Tahany Emad	Game Design & Content	Designs the trade-off model, events and balancing, and writes age-level content and debriefs.
Aml Ashraf	Visuals & UI/UX	Designs the outpost dashboard, HUD and planet looks, including Arabic layout.
Rawan Hany	QA, Docs & Media	Runs playtests (including with children 7–12), writes documentation and data attribution, and produces the demo video.
Alaa Elsayed	Gameplay Developer	Implements the Outpost Command loop, resource logic and events, and connects the dashboard to the data and assistant.

Workflow Strategy

Workflow: problem analysis → trade-off design → build → data and AI integration → playtesting → demo preparation.

Collaboration: a shared GitHub repository with small, frequent commits, one group chat for quick decisions, and a short sync at the start of each work block. Each member owns one area, and the AI and gameplay owners review each other's integration points.

Milestones (13–14 November)

- **Day 1, morning:** lock the trade-off model and resources; connect to DONKI and prepare the archived fallback.
- **Day 1, afternoon:** playable outpost loop with events and the dashboard.
- **Day 2, morning:** source attribution, debriefs, assistant knowledge; playtest in English and Arabic; fix bugs.
- **Day 2, afternoon:** record the demo video, finish the README and data explanation, and submit.

Resources

Software: Python 3.12, Ursina Engine, Google Gemini API (google-genai), ChromaDB, SQLite, arabic-reshaper, python-bidi, pytest, GitHub.

Hardware: standard laptops.

NASA data: DONKI space-weather database (api.nasa.gov); Open Science Data Repository, OSDR (osdr.nasa.gov); NASA Moon, Mars, Europa and ISS research pages (science.nasa.gov/moon/facts, science.nasa.gov/mars/facts, science.nasa.gov/jupiter/jupiter-moons/europa/facts, nasa.gov/international-space-station).

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Team

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